

DANGOTE CEMENT PLC — Digital KPI Automation: Business Case & Supporting Docs

1. Business Case

Manual KPI reporting across 4 plants (Obajana, Ibese, Gboko, Okpella; 17.7 Mt/yr, 59.6% EBITDA margin) causes delays, data errors, and weak governance on a N3.3T revenue base. Proposed solution: Power Platform stack (Power Apps + SharePoint + Power Automate + Dataverse + Power BI) with two-tier storage (SharePoint entry, Dataverse reporting).

Item	Value	Notes
Initial Investment	N8.5M	Dev, 1st-yr licences, training, change mgmt
Annual Licence (run-rate)	N3.2M	Power Apps Premium (20) + Power BI Pro (15)
Annual Savings Target	N45–70M	Admin time, faster decisions, fewer incidents
Payback Period	2–4 months	From operator time savings + error reduction
3-Year Net Benefit	N110–180M	After licence and maintenance costs
EBITDA Impact	+0.3–0.5%	From reduced wastage, better OEE tracking

Key assumptions: 30 min/operator/day saved (N400/hr loaded); 10% off-spec reduction = ~N15M/yr/plant; 1–2% OEE gain within 6 months from same-day KPI visibility. Recommendation: 8-week Obajana pilot (N2.8M) → full rollout decision at Month 3.

2. Prerequisites & Assumptions

Requires: M365 tenant (SharePoint/Power Apps/Automate/BI), Standard E3/E5 licences for all users, Power Apps Premium (~20 seats), Power BI Pro (~15), Dataverse provisioned, 5 SharePoint lists, plant champion per site, M365 admin access. Assumptions: standard M365 infra deployable; operators have basic digital literacy; 10:00 AM daily deadline; 4 plants + 12 kiln lines as independent units; adequate internet at plant sites. Data limitations: synthetic prototype data; thresholds need pilot calibration; schema assumes one shift per line per day.

3. Implementation Roadmap

0–30 Days (Pilot): Deploy Power Apps + 5 SP lists + Power Automate flows + Power BI RLS at Obajana. Train operators. 2-week parallel run with Excel. Compare quality/timeliness.

1–3 Months (Rollout): Scale to Ibese, Gboko, Okpella (weeks 5–10). Train all operators. Refine rules from pilot data. Configure DQO/Regional Manager views. Retire Excel.

3–6 Months (Optimise): Add predictive trend alerts. ERP integration. SCADA for kiln feed & power capture. Executive benchmarking dashboard. Quarterly rule review cycle.

Resource	Role	Effort
Power Platform Developer	Setup, config, deployment, integration	2 wk + 1 wk/plant
Plant Champion	Power user & peer trainer per site	~2 hrs/wk ongoing
IT Administrator	Tenant config, DLP, RLS setup	1–2 wks total
Licensing	M365 E3/E5 + PP Premium (20) + BI Pro (15)	N3.2M/yr

4. Risk & Safety Assessment

#	Risk	L	I	R	Mitigation	Owner
1	RLS misconfig → data exposure	L	H	M	Test 5 roles pre-launch; quarterly RLS audit	IT Admin
2	Operator resistance to new system	M	M	M	Phased rollout + champions; 2-wk parallel run	Plant Mgr
3	Data entry errors bypass validation	M	M	M	Multi-layer validation + sign-off + DQO oversight	DQO
4	SP list threshold (5K items) hit	M	L	L	Indexed columns, filtered views, quarterly archiving	IT Admin
5	Power Platform outage	L	H	M	Offline mode; 4-hr restore SLA; manual backup	IT Admin
6	Data loss in SP→Dataverse promotion	L	H	M	FACT_ValidationLog audit trail; transactional rollback	IT Admin
7	Inadequate training → poor data	M	H	H	Structured training; train-the-trainer; reference guides	Champion
8	Licence cost overrun	M	L	L	Std M365 for operators; 20-seat buffer estimate	Finance
9	SCADA/ERP integration delay	M	M	M	Core system independent; integration is additive	Developer

Safety: Process — automated quality alerts + NCR flags prevent off-spec dispatch. Data — RLS, no Excel handling, immutable audit trail, M365/TLS encryption. Operational — low-code admin, no new hardware, offline-capable, paperless.